

2.2.2 -

- a. $p \wedge q$
- b. $p \wedge \neg q$
- c. $\neg(p \vee q)$
- d. $r \wedge (p \vee q)$
- e. $(r \wedge p) \vee q$
- f. $(r \wedge p) \wedge \neg q$

2.2.3 -

- a. T
- b. T
- c. T
- d. F
- e. F
- f. F

2.2.4

b.

p	q	$p \vee q$	$\neg(p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

c.

P	Q	R	$\neg Q$	$P \wedge \neg Q$	$R \vee (P \wedge \neg Q)$
T	T	T	F	F	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	F	T
F	T	F	F	F	F
F	F	T	T	F	T
F	F	F	T	F	F

2.2.5 -

- a. I am not going to a movie tonight
- b. I am going to a movie and the party tonight
- c. I am going to a movie and not the party tonight

- d. I am not going to the movie or the party tonight
 e. I am not going to the movie and the party tonight (Clearly I have enough homework to do)

2.2.7 -

- a. B or D or M
 b. (B and D) or (D and M) or (B and M)
 c. B or (D and M)

2.2.8 -

- a. $r = F, q = T, p = F$
 b. $p = T, q = T$
 c. $p = T, q = T$

2.3.1 - Do not need to explain why each statement is t/f

- a. False, feb has 28-29 days
 b. False, 7 is not an even number
 c. True, 7 is odd and February does not have 30 days
 d. False, January has 31 days

2.3.3 - Six things for each part

- a. Inverse: if 3 is not a prime, then 5 is not even (true)
 Contrapositive: If 5 is not even, then 3 is not a prime (false)
 Converse: if 5 is even, then 3 is a prime (true)
 b. Inverse: If 7 is greater than 5, then 5 is greater than 3 (true)
 Contrapositive: If 5 is greater than 3, then 7 is is greater than 5 (true)
 Converse: If 5 is less than 3, then 7 is less than 5 (false)
 c. Inverse: If 3 is a positive number, then 5 is a negative number (false)
 Contrapositive If 3 is a negative number, then 5 is a positive number (false)
 Converse: If 3 is a positive number, then 5 is a negative number (false)

2.3.4 -

a.

p	q	$\neg p$	$\neg p \wedge q$	$(\neg p \wedge q) \rightarrow p$
T	T	F	F	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

c.

p	q	$\neg p$	$\neg p \vee q$	$q \rightarrow p$	$(p \vee q) \leftrightarrow (q \rightarrow \neg p)$
T	T	F	T	F	F
T	F	F	T	T	T
F	T	T	T	T	T
F	F	T	F	T	F

d.

2.3.5 -

- a. Not j, therefore c
- b. if c, therefore not j
- c. not j, c
- d. c if not j
- e. not c
- f. j and c

2.3.8 - Translate each proposition into english exactly as it is stated

- a. The roads were wet therefore traffic was heavy
- b. The roads were wet and there was an accident!
- c. There was not an accident and heavy traffic
- d. The traffic was heavy, therefore the roads were wet and there was an accident
- e. The roads were wet and there was light traffic

2.3.9 -

- a. Not y, p
- b. if (y or p) therefore c
- c. c if p
- d. if p = t, therefore c

2.3.10 -

- a. unknown
- b. true
- c. false
- d. false
- e. unknown
- f. false

2.4.1 - For each part, give a truth table and explain how it justifies your answer

- a. Tautology
- c. Neither
- e. Contradiction

2.4.4 - For each part, give a truth table and explain how it justifies your answer

- a. Yes, they are logically equivalent
- c. Yes, because of the idempotent law, the p's can combine

2.4.6 - Note that there are four things to do for each part of this ex.

- a.
- b.

2.5.2 - Do not skip any steps in proofs. For full credit, indicate which law used at each step of each proof of logical equivalence

- a.
- b.
- c.
- g.

2.5.3 - Same as above

- c.
- e.

2.5.4 - First, state the assertion as a statement of logical equivalence, or not, of two expressions. Second, give a counterexample or a proof for the assertion, as appropriate.

- a.
- b.
- c.

I didn't realize this homework was due thursday and not friday, otherwise I would have spent more time completing it.

Inverse: If I spent more time completing it, then I would have known the homework was due thursday and not friday

Converse: If I did not know the homework was due thursday and not friday, then I would hnot have spent more time completing it

Contrapositive: If i did not spend more time completing it, then I did not know the homework was due thursday and not friday.

That's a little joke for ya.